# OpenAlex as the Literature Database of Choice of IPBES

**White Paper**

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**GitHub Repo: https://github.com/IPBES-Data/IPBES_WP_OpenAlex**

## Glossary

- API: Application Programming Interface. An interface to query the database programmatically.
- data snapshot: A snapshot of the data which can be downloaded for local use.

## Introduction

This white paper evaluates OpenAlex as a recommended data source for IPBES assessments of scholarly literature, compared to Web of Science and Scopus. The focus is on its use within IPBES, emphasizing the transparency of academic literature searches.

In IPBES assessments, the main sources for scholary literature search were Web of Science and Scopus. These were mainly used via the web interface due to licensing limitations and complexities in the API. The use of the web interface poses a considerable problem regarding transparency and reproducibility of the search. In addition, neither the data used of these scholary data sources nor the code which is used to conduct the search or the API implementation is openly availabe. As literature forms the foundation of most aspects in IPBES assessments, this lack of transparency filters through to the final product. The Data and Knowledge Management Policy (https://www.ipbes.net/node/38260 and https://zenodo.org/records/6243095) aims to enhance the transparency of IPBES products. An option to the replace the proprietary scholarly data sources is OpenAlex. Unlike the

proprietary and restricted scholarly data sources, OpenAlex is open, free to use, and both the data and code are openly accessible under the CC0 license.

The data tsu therefore decided to assess the suitability of OpenAlex as a replacement for the proprietary scholary data sources in the context of IPBES, with particular focus on the needs of the assessments. Based on this initial assessment and the initial usage in assessments, we recommend the usage of OpenAlex instead of proprietary scholary data sources.

This document will detail differences between these sholary data sources and outline the advantages of OpenAlex over WoS and Scopus.

## Advantages OpenAlex in comparison to other databases

The comparison needs to be done in several aspects which are:

- Number of works in the dta source and number of OpenAccess works
- Coverage (e.g Global South, academic, grey literature, etc)
- Openness of data and code used
- Web and API Access
- costs for usage

We will go throuh these points in more detail now.

### Available Works and Number of OpenAccess works

OpenAlex was seeded with data from Microsoft Academic when it was retired in 2021. The data was then enriched with data from Crossref and is continuously updated from several sources (See here).

As can be seen in this table, OpenAlex has more works than any other scholary data source with the exception of google scholar.

Most works in OpenAlex are ingested via Crossref, but a large proportion are from other sources.

| | Number of works | Open Access Works | Citations | Price | Data Openness | Org Structure |
|---|---|---|---|---|---|---|
| OpenAlex | 243M | 48M | 1.9B | Freemium | Fully open, CC0 license | Non-profit |
| Scopus | 87M | 20.5M (ref) | 1.8B | Subscription | Closed | For Profit |
| Web of Science (core) | 87M (ref) | 12M (ref) | 1.8B | Subscription | Closed | For Profit |

| | Number of works | Open Access Works | Citations | Price | Data Openness | Org Structure |
|---|---|---|---|---|---|---|
| Dimensions | 135M | 29M (ref) | 1.7B | Freemium | Partly open, personal use | For Profit |
| Google Scholar | 389M (estimated) | ? | ? | Free | Closed | For Profit |
| Crossref | 145M | 20M | 1.45B | Free | Fully open, CC0 license | Non-profit |

(From: OpenAlex)

Culbert et al. (2024) analysed the coverage of OpenAlex, Web of Science and Scopus and came to the conclusion that

> [...] when restricted to a cleaned dataset of 16,788,282 recent publications shared by all three databases, OpenAlex has average reference numbers comparable to both Web of Science and Scopus.

### Coverage (e.g Global South, academic, grey literature, etc)

OpenAlex has a broad coverage from the Global South. It also has a large proportion of non-academic works as well as works without DOIs, i.e. including reports and data.

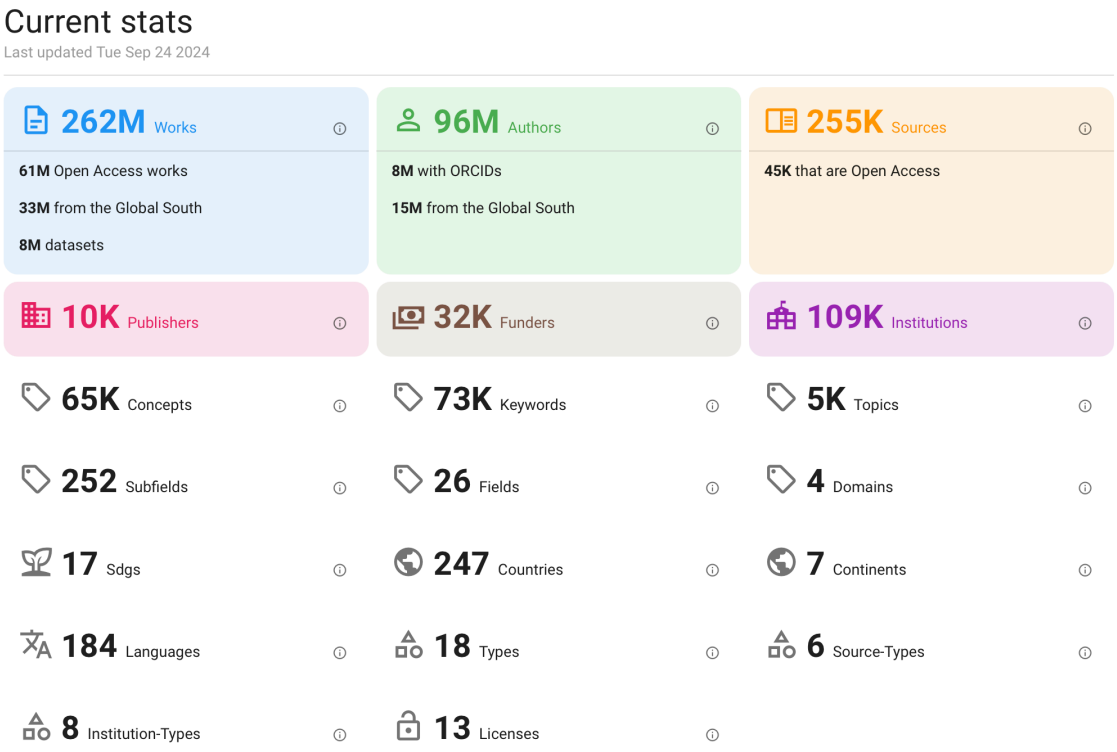


Figure 1: From: OpenAlex

Alonso-Alvarez & van Eck analysed the coverage and metadata of African publications in particular, and came to the conclusion, that

> Our analysis shows that OpenAlex offers the most extensive publication coverage. In terms of metadata, OpenAlex offers a high coverage of publication and author information. It performs worse regarding affiliations, references, and funder information. Importantly, our results also show that metadata availability in OpenAlex is better for publications that are also indexed in Scopus or WoS.

### Openness of Data and Code used

OpenAlex is free and open, i.e. it is free to use for everybody without any subscription or registration, by design. The data is freely available by downloading a data snapshot by following the instructions on OpenAlex. The datasets can therefore be analysed locally using approaches which are not provisioned for in the API.

### Web and API Access

API access is important for accessing the database programmatically. An example is a search for all articles covering a specific topic. Usually, this is an iterative provess which needs refinement of the search terms. The traditional way is to type the search term in the search box and and write down at least the number of hits, change the search term and repeat the process. Especially when other comparisons should also be done (e.g. assessment of certain key-papers are in the search results), this becomes a very tedious and error prone process.

This process can be automated by using the API. In this case, the search is defined e.g. in R or any other programming language and the search is done programmatically. The results can then be analysed and the search terms can be refined in a reproducible way with transparent criteria. The results can be shown in a clear, concise and reproducible automatically generated report.

Nevertheless, web acces to conduct manual searches is a valuable tool to conduct initial testing of search terms. render OpenAlex has, as all of the other other scholary data sources, a web interface.

In addition to the web interface, most scholary sources, with the notable exception of google scholar, have an API which can be used to access the data programmatically.

In contrast to the APIs assessed, (WoS, Scopus), the OpenAlex API is relatively simple but provides all the neccessary functionality needed for all IPBES related tasks. For more complex searches or analysis, a data snapshot can be stored and analysed locally.

To simplify the API usage even more, the excellent R package openalexR is available to access the API from R.

### Costs

OpenAlex is free to use, which is not the case for most other databases. If the limit for free usage (100k API calls per day with a maximum of 10 per second) is not enough, a Premium subscription can be obtained (the tsu has obtained a Premium license free of charge due to extensive feed back and extensive discussions with OpenAlex). The pricing is determined on a case by case basis.

For nearly all usage scenarios within IPBES, the free tier is more than enough, and the remaining usage scenarios are still possible but require longer, or can utilise a data snapshot.

In contrast, the API costs for WoS can easily go into 10s of thousands per dollars per year, and if a data snapshot is required even in the millions. In addition, none of the offered smaller API licenses covers the needs of IPBES, i.e. download of diverse metadata as well as download of a large nuber of works.

## Experience using OpenAlex

OpenAlex was used in the Transformative Change Assessment to fulfil several requests from the experts. The use of OpenAlex was suggested by the d&k tsu to the experts and the experts decided to use OpenAlex. The main reason was that the analysis and search they had planned would have been essentially not possible with scopus or WoS due to limitations of the GUI and non-availability (and limitations) of the APIs.

OpenAlex enabled us to compile and download the Transformative Change Corpus (TCA Corpus), consisting of 4.5 million works. The works were downloaded, searches in the TCA Corpus were done for several aspects of the TCA and evidence was produced which was considered by the authors as very important to underline their research as well as show venues for additional focal points within the assessment as well as further research based on the assessment.

OpenAlex also enabled us to do a second kind of search, i.e. a snowball search (following the citation graph from selected key papers one genertion back and one generation forward). This very focussed search is easily possible in OpenAlex via the API and the respective R package (openalexR), and has been used in the TCA as well as several times in the Business and Biodiversity Asserssment to obtain corpi of focussed literature to, im most cases, find additional evidence for certain points. It is also in usage for the Monitoring Assessment, which is at the moment conduting at least one snowball and one larger keyword search using OpenAlex.

Overall, the experience with using OpenAlex within IPBES has been very positive and, as multiple requests from outside IPBES indicate, on the front of literature search methodologies in general.

## Conclusion

We therefore recommend OpeAlex as the preferred scholary data sources for IPBES as it is a suitable, and even more powerful replacement for WoS and Scopus.

To facilitate the switch, the following technical guidelines are available and in preparation / planning to help the authors to use OpenAlex programmatically in their assessments

- **Snowball Search for Literature Search and Analysis** completed
- **Keyword based Literature Search and Analysis using OpenAlex** in planning
- **Using OpenAlex data snapshots - download, storage and querying and analysing** early planning stages